Lesson plan · for the teacher

Teaching and assessing code: unit check

Programming · Level 3 · Unit check

Time	Materials
45 min	Board, chalk, paper. No devices needed.

Learning goals

- Read before you write: PRIMM
- Parsons problems
- Teaching debugging
- Misconceptions and feedback

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. For each wrong option, name the misconception behind it
4. Write the hardest question on the board for tomorrow

Discussion

- Which technique will you use first, and with which class?
- Which question surprised you, and why?
- Which misconception will you look for next week?

Exit ticket

- One change I will make to how I teach programming

Try it in class tomorrow

Pick one technique from this course (a PRIMM predict question, Parsons strips, the debugging routine or a hinge question) and use it in your next programming lesson. Then tell a colleague what the pupils' answers showed.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Which PRIMM stage is "Change the program so it counts in twos"?**

◯ Predict ◯ Investigate ◯ Modify ◯ Make

2 **A good Parsons problem has exactly one correct order.**

◯ True ◯ False

3 **Put the lines in order: the program prints the class average. One line doesn't belong.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

☐ `average = total / len(marks)`

☐ `marks = [12, 15, 9]`

☐ `average = total / 2`

☐ `total = sum(marks)`

☐ `print("Average:", average)`

4 **This program should print 1 to 10. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  # print 1 to 10
2  for n in range(1, 10):
3      print(n)
```

5 **Match each idea to what the research says.**

PRIMM	• • • •	A driver and a navigator swap roles often
Parsons problems	• • • •	Each wrong answer points to a misconception
Hinge questions	• • • •	Ordering lines takes less time than writing, with similar learning
Pair programming	• • • •	Predict and run before you modify and make